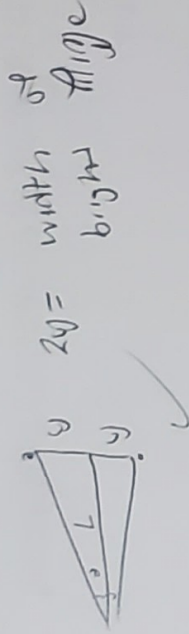


Name: _____

Date: _____

3. How wide is the central diffraction peak on a screen 2.30 m behind a 0.0348 mm wide single slit illuminated by 558 nm light? Show all your work.

[3 marks]



$$\sin \theta = m \frac{\lambda}{a}$$

$$\theta = \sin^{-1} \left(\frac{\lambda}{a} \right)$$

$$\theta = \sin^{-1} \left(\frac{558 \times 10^{-9}}{0.0348 \times 10^{-3}} \right)$$

$$\theta = 0.9187^\circ$$

$$\tan(\theta) \cdot L = y$$

$$0.00868 \text{ m} = y$$

$$0.01737 \text{ m} = 2y$$

or fringe

so it's a width of

$$0.01737 \text{ m}$$

Final answer: 0.01737 m

4. The first-order line of 589 nm light falling on a diffraction grating is observed at a 14.5° angle.

a) How far apart are the slits? Show all your work.

b) At what angle will the third order be observed? Show all your work.

a) $\sin \theta = m \frac{\lambda}{d}$ $m=1$

$$\sin \theta = \frac{589 \times 10^{-9}}{d}$$

$$d = \frac{589 \times 10^{-9}}{\sin 14.5^\circ}$$

$$d = 2.352 \times 10^{-6} \text{ m apart}$$

[4 marks]

b) $\sin \theta = m \frac{\lambda}{d}$ $m=3$ third order

$$\sin \theta = 3 \frac{\lambda}{d}$$

$$\theta = \sin^{-1} \left(3 \frac{\lambda}{d} \right)$$

$$\theta = 48.69^\circ$$

Final answer for a: $2.352 \times 10^{-6} \text{ m}$

Final answer for b: 48.69°

Name: _____

Date: _____

APPLICATION

1. Pulsed lasers used for science and medicine produce very brief bursts of electromagnetic energy. If the laser light wavelength is 1062 nm (Neodymium-YAG laser), and the pulse lasts for 34 picoseconds (34×10^{-15} s).

peta seconds 10^{15}

[3 marks]

- a) How many wavelengths are found within the laser pulse? Show all your work.
 b) How brief would the pulse need to be measured in seconds to fit only one wavelength?
 Show all your work.

$$a) \lambda = 1062 \text{ nm} \quad \text{pulse length} = 34 \times 10^{-15} \text{ s}$$

$$\frac{\text{dist}}{\lambda} = \text{wavelengths in beam} \quad d = v \cdot t$$

$$d = (3 \times 10^8 \text{ m/s} \times 34 \times 10^{-15} \text{ s})$$

$$d = 1.02 \times 10^{-6} \text{ m}$$

$$3.54 \times 10^{-15} \text{ s}$$

9 wavelengths can be found

$$\lambda = 9.6$$

Final answer for a: 9 full wavelengths

Final answer for b: $3.54 \times 10^{-15} \text{ s}$
to make one wavelength

$$f \lambda = c$$

$$b) f = \frac{\text{cycles}}{\text{second}} \quad T = 2.82 \times 10^{14} \text{ Hz}$$

$$\frac{1}{T} = \text{seconds/cycle} \quad 3.54 \times 10^{-15} \text{ s per } \lambda$$

2. A distant galaxy emits light that has a wavelength of 572.6 nm. On earth, the wavelength of this light is measured to be 573.1 nm.

 $2 \frac{1}{2}$ meters

[3 marks]

- a) Does this galaxy approaches or recedes from the earth. Explain.
 b) Find the speed of the galaxy relative to the earth. Show all your work.

$$f_o = f_s \left(1 \pm \frac{v}{c}\right)$$

a) approaching because the frequency increased relative to the source, which indicates a relative velocity towards each other

$$\frac{573.1}{572.6} - 1 = \frac{v}{c}$$

Since $\lambda_{\text{earth}} > \lambda_{\text{galaxy}}$, then $v_{\text{galaxy}} < + \text{velocity}$

$$\left(\frac{573.1}{572.6} - 1\right) c = v$$

$$2619629 \text{ m/s} = v$$

See

Solutions!

galaxy moving away from earth

Final answer for a: approaching earth

Final answer for b: 261962.98 m/s

Name: _____

Date: _____

COMMUNICATION

1. Can electromagnetic waves travel through a perfect vacuum? Can sound waves through a perfect vacuum? Explain why or why not.

[2 marks]

Yes they can, electromagnetic waves do not require a

medium to travel to since both the electric and magnetic components of the wave do not require a medium to have an influence, so the wave propagates itself since it depends on the electric and magnetic fields. Sound waves cannot since they are differences in pressure, thus by definition they need a medium to propagate through.

2. i) Is polarization a property of all electromagnetic waves, or is it unique to visible light? ^{no} yes

ii) Can sound waves be polarized?

iii) What does polarization tell us about the nature of light? transverse

[3 marks]

i) Yes it is, since visible light is a transverse wave.

ii) No, sound waves cannot be polarized since they are longitudinal waves, so a slit won't do anything.

iii) It tells us that light is a transverse wave since only transverse waves can be polarized. (note: particles in transverse waves move perpendicular to the direction of wave propagation.)

3. If Young's double-slit experiment were submerged in water, how would the fringe pattern be changed?

[1 mark]

The fringe pattern would change since λ changes

$$\sin \theta = \frac{\lambda}{d}$$

$\frac{\lambda_{\text{vac}}}{n} = \lambda_{\text{water}}$ Since $\lambda_{\text{water}} < \lambda_{\text{vac}}$, the fringes would move closer together due to the smaller angle θ .

THINKING/INQUIRY

Red laser from the physics lab is marked as producing 632.8 nm light. When light from this laser falls on two closely spaced slits, an interference pattern formed on a wall several meters away has bright red fringes spaced 5.00 mm apart near the center of the pattern. When the laser is replaced by a small laser pointer, the fringes are 5.14 mm apart. What is the wavelength of light produced by the laser pointer?

Show all your work.

$$\lambda = 632.8 \times 10^{-9} \text{ m} \quad 5 \times 10^{-3} \text{ m from center}$$

$$5.14 \times 10^{-3} \text{ m from center}$$

[3 marks]

$$\sin \theta = m \frac{\lambda}{d}$$

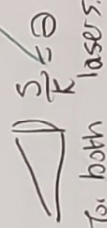
$$\sin \theta = \frac{\lambda}{d}$$

$$\left(\tan^{-1} \left(\frac{y}{L} \right) \right) = \frac{\lambda}{d}$$

$$\frac{y}{L} = \frac{\lambda}{d}$$

$$= \frac{L}{L}$$

$$m=1 \text{ Assumes } \theta \approx \tan \theta \approx \sin \theta$$



same for both lasers.

$$\frac{y}{\lambda_1} = \frac{y}{\lambda_2}$$

$$\frac{y}{\lambda_1} = \frac{y}{\lambda_2}$$

$$\frac{y}{\lambda_1} = \frac{y}{\lambda_2}$$



$$632.8 \times 10^{-9} \cdot \frac{5.14 \times 10^{-3}}{5 \times 10^{-3}} = \lambda_2$$

$$650 \times 10^{-9} = \lambda_2$$

Final answer: $650.51 \times 10^{-9} \text{ m}$

Monochromatic light falls on two very narrow slits 0.048 mm apart. Successive fringes on a screen 6.50 m away are 8.5 cm apart near the center of the pattern. Determine the wavelength and frequency of the light.

Show all your work.

$$0.048 \times 10^{-3} \text{ m} = d$$

$$\frac{y}{L} = \frac{\lambda}{d}$$

$$\sin \theta = m \frac{\lambda}{d}$$

$$\theta = \tan^{-1} \left(\frac{y}{L} \right)$$

$$\theta = 0.3746^\circ$$

$$\sin \theta = \frac{\lambda}{d}$$

$$\sin \theta d = \lambda$$

$$(0.048 \times 10^{-3}) = \lambda$$

$$3.13 \times 10^{-7} = \lambda$$

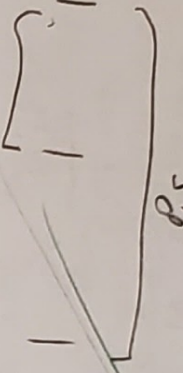
$$313.8 \text{ nm} = \lambda$$

$$c = \lambda f$$

$$\frac{c}{\lambda} = f$$

$$3.0 \times 10^8 \text{ m/s} = f$$

$$m=1$$



[4 marks]

LOOK ON PAPER

Final answer: wavelength

frequency

313.8 nm

$9.551 \times 10^{14} \text{ Hz}$

answers of 1. B. C. D. E. F. G. H. I. J. K. L. M. N. O. P. Q. R. S. T. U. V. W. X. Y. Z.

OK